

Chapter 17 – Gravitation

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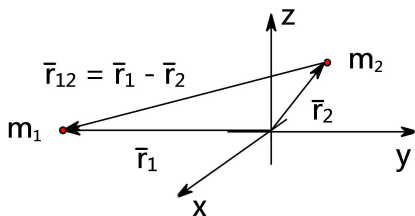
Newton's Law of Gravitation

Features of the gravitational interaction

- source: any mass (matter)
- one of the four fundamental interactions
- relative strength: much weaker than the other three interactions (example: two electrons)
- long-range; important for the evolution of the Universe
- always attractive

Newton's Law of Gravitation

Force of gravitational attraction between two particles
(particle = mass concentrated at a single point of space)



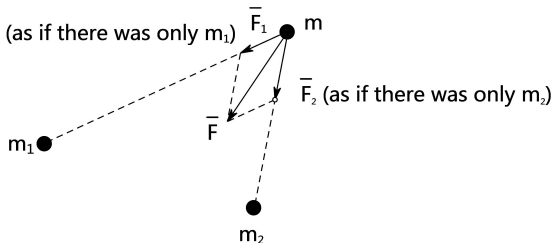
$$\underbrace{\vec{F}_{12}}_{\text{on "1" due to "2"}} = -G \frac{m_1 m_2}{r_{12}^2} \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$$

$$G = 6.67250(85) \cdot 10^{-11} \left[\text{N} \cdot \frac{\text{m}^2}{\text{kg}^2} \right]$$

Gravitational constant

Note. $\vec{F}_{12} = -\vec{F}_{21}$

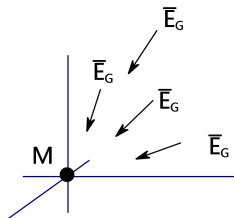
The superposition principle



Gravitational Field

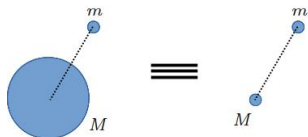
The gravitational interaction defines a vector field in space

$$\vec{E}_G = \frac{\vec{F}_G}{m} = -G \frac{M}{r^2} \frac{\vec{r}}{r}$$



Fact (will be justified soon)

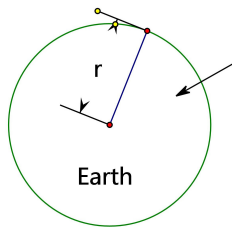
For a spherically symmetric distribution of mass in a region Ω , the gravitational field outside of Ω is as if the whole mass was concentrated at the center of Ω .



Weight

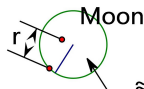
The weight of a body is the total gravitational force exerted on a body by all other objects in the Universe.

In practice,



$\approx$ only due to earth

$$F \approx G \frac{M_{\text{earth}} m}{r^2}$$



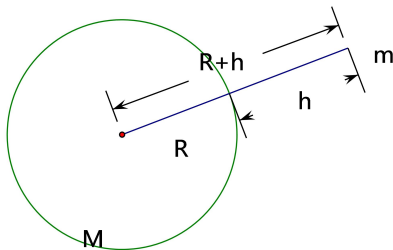
$\approx$ only due to moon

$$F \approx G \frac{M_{\text{moon}} m}{r^2}$$

At the Earth's surface, $F = G \frac{M_{\text{Earth}} m}{R^2} = mg$, where R is the radius of the Earth.

Note. By measuring g , it is possible to estimate the mass of the Earth as $M_{\text{Earth}} = gR^2/G \approx 5.98 \times 10^{24}$ kg. Hence, the average density of the Earth's mass $\rho_{\text{av}} = M_{\text{Earth}} / \frac{4}{3}\pi R^3 \approx 5.5 \times 10^3$ kg/m³.

Variation of the weight with altitude (close to the Earth)



$$\begin{aligned}\text{weight} &= G \frac{Mm}{(R+h)^2} = G \frac{Mm}{R^2(1+\frac{h}{R})^2} = G \frac{M}{R^2} m \left(1 + \frac{h}{R}\right)^{-2} \\ &= G \frac{M}{R^2} m \left(1 - 2\frac{h}{R} + 3\left(\frac{h}{R}\right)^2 - \dots\right) \quad [\text{and, if } h \ll R] \\ &\approx \underbrace{G \frac{M}{R^2}}_{=g} m = mg \quad (\text{close the Earth's surface})\end{aligned}$$

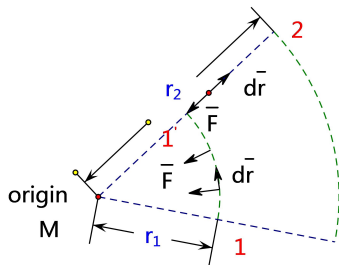
Gravitational Potential Energy

Gravitational Potential Energy

A force $\vec{F}$ is called *central*, if $\vec{F} = f(r)\vec{r}$. Example: the gravitational force due to a point mass placed at the origin

$$\vec{F} = -G \frac{Mm}{r^2} \frac{\vec{r}}{|\vec{r}|} = f(r)\vec{r}, \text{ with } f(r) = -G \frac{Mm}{r^3}.$$

Fact. Central forces are conservative $\implies$ their work does not depend on the path.



Choose the path:

1 $\rightarrow$ arc of a circle $\rightarrow$

1' $\rightarrow$ radius $\rightarrow$ 2'

$$W_{1 \rightarrow 2} = \int_{1 \rightarrow 2} \vec{F} \circ d\vec{r} = \underbrace{\int_{1 \rightarrow 1'} \vec{F} \circ d\vec{r}}_{=0 \quad (\vec{F} \perp d\vec{r})} + \underbrace{\int_{1' \rightarrow 2'} \vec{F} \circ d\vec{r}}_{\angle(\vec{F}, d\vec{r}) = \pi} = - \int_{r_1}^{r_2} |\vec{F}| \cdot |d\vec{r}| =$$

$$= - \int_{r_1}^{r_2} G \frac{Mm}{r^2} dr = G \frac{Mm}{r} \Big|_{r_1}^{r_2} = G \frac{Mm}{r_2} - G \frac{Mm}{r_1}$$

On the other hand, for potential forces, $W_{1 \rightarrow 2} = U_1 - U_2$. Hence, the gravitational potential energy

$$U(r) = -G \frac{Mm}{r} + C$$

the additive constant C can be chosen arbitrarily (no physical meaning) as only ΔU is measurable/physical.

Note. $\vec{F} = -\text{grad } U = -\nabla U$.

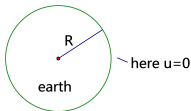
Choice of C ("choice of the gauge") — infinitely many possibilities. Two examples:

① $U(\infty) = 0 \implies C = 0$ and

$$U(r) = -G \frac{Mm}{r}$$

② $U(R) = 0 \implies -G \frac{Mm}{R} + C = 0 \implies C = G \frac{Mm}{R}$ and

$$U(r) = GMm \left(\frac{1}{R} - \frac{1}{r} \right)$$

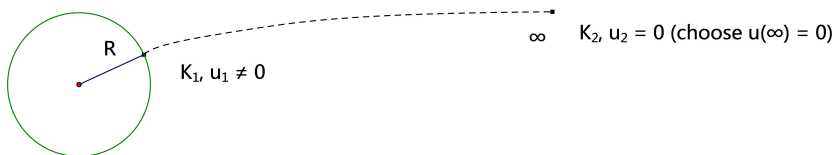


Let $r = R + h$ (with $h \ll R$), then

$$\begin{aligned} U(r) &= GMm \left(\frac{1}{R} - \frac{1}{R+h} \right) = G \frac{Mm}{R} \left(1 - \frac{1}{1 + \frac{h}{R}} \right) \\ &= G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} - \left(\frac{h}{R} \right)^2 + \dots \right) \quad [\text{for } h \ll R] \\ &\approx G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} \right) = G \frac{Mm}{R^2} h = mgh \end{aligned}$$

Example. Escape Speed

Escape speed — the minimum speed a particle should have to be able to move away from a planet to ∞ .



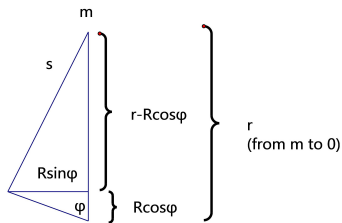
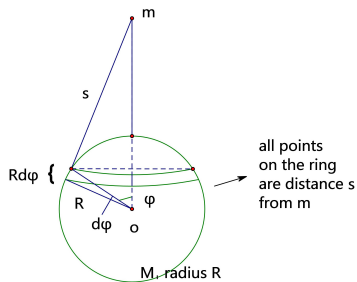
Conservation of energy

$$K_1 + U_1 = K_2 + U_2$$

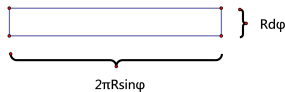
Hence, e.g. for the Earth,

$$\frac{1}{2}mv_{\text{esc}}^2 - G\frac{M_{\text{Earth}}m}{R} = 0 \implies v_{\text{esc}} = \sqrt{\frac{2GM_{\text{Earth}}}{R}} \approx 11.2\text{km/s}$$

Example. Potential Energy Due to a Uniform Spherical Shell



Area of the infinitesimal ring $dA = 2\pi R \sin \varphi R d\varphi$



Mass on the ring $dM = \frac{M}{4\pi R^2} dA = \frac{1}{2} M \sin \varphi d\varphi$. Contribution of the ring to the potential energy (choose gauge with $U(\infty) = 0$)

$$dU = -G \frac{m dM}{s} = -G \frac{mM}{2s} \sin \varphi d\varphi$$

Now, express everything in terms of s ; then integrate w.r.t. s from $r - R$ ("north pole") to $r + R$ ("south pole")

$$s^2 = (r - R \cos \varphi)^2 + (R \sin \varphi)^2 = r^2 - 2rR \cos \varphi + R^2$$

$$2s \, ds = 2rR \sin \varphi \, d\varphi \implies \sin \varphi \, d\varphi = \frac{s}{Rr} \, ds$$

Hence,

$$dU = -G \frac{Mm}{2s} \frac{s}{Rr} \, ds = -G \frac{Mm}{2Rr} \, ds$$

and,

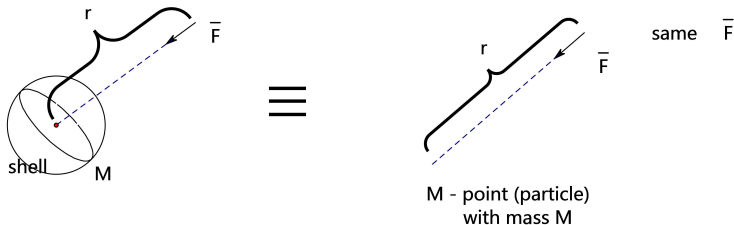
$$\begin{aligned} U(r) &= \int_{r-R}^{r+R} \left(-G \frac{Mm}{2Rr}\right) ds = -G \frac{Mm}{2Rr} [r + R - r + R] \\ &= -G \frac{Mm}{r} \implies \boxed{U(r) = -G \frac{Mm}{r}} \end{aligned}$$

The corresponding force

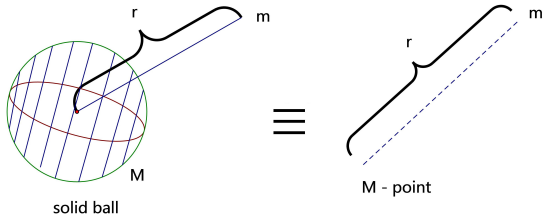
$$\vec{F} = -\text{grad } U = -G \frac{Mm}{r^2} \frac{\vec{r}}{r}$$

Discussion of the result

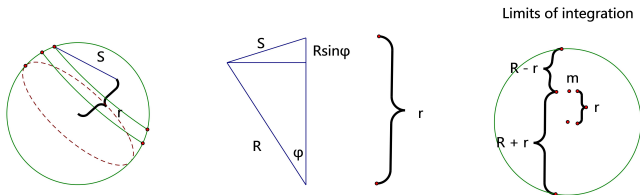
(*)



(*) For a solid ball with a spherically symmetric distribution of mass, divide it into shells and use the above fact



(*) If m is inside the sphere

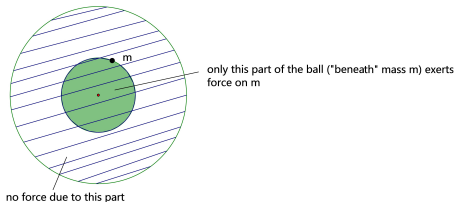


$$\boxed{U(r)} = -G \frac{mM}{2rR} \int_{R-r}^{R+r} ds = \boxed{-G \frac{Mm}{R} = \text{const}}$$

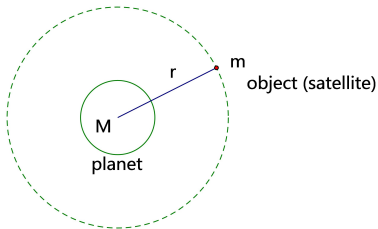
$$\Rightarrow \vec{F} = -\text{grad } U \equiv 0$$

Gravitational force on mass m inside the sphere is zero!

(*) Consequently



Motion of Satellites in Circular Orbits



Gravitational force plays the role of the centripetal force

$$\vec{F}_{grav} = m\vec{a}_{centripetal}$$

Magnitude

$$G \frac{Mm}{r^2} = \frac{mv^2}{r} \quad (\text{or} \quad m\omega^2 r)$$

Satellite's linear speed in a circular orbit $v = \sqrt{\frac{GM}{r}}$.

Note. This value (if $r = R$, i.e. for low orbits) is exactly $\sqrt{2}$ times smaller than v_{esc} .

Period of motion in a circular orbit

$$\boxed{T} = \frac{2\pi r}{v} = 2\pi \frac{r^{3/2}}{\sqrt{GM}} \boxed{\sim r^{3/2}}.$$

Given the orbit's radius r , the speed is determined. Note that further orbits mean slower speeds ($v \sim 1/\sqrt{r}$) and longer periods.

Examples

(a) geostationary satellite ($T_{geost} = 24h$)

$$r_{geost} \simeq 35786 \text{ km}$$

$$v_{geost} \simeq 3.1 \text{ km/s}$$

(b) ISS ($r = 6800 \text{ km}$)

$$v = 7.7 \text{ km/s}$$

$$T = 93 \text{ min}$$

(c) moon ($r \simeq 380000 \text{ km}$)

$$v = 1 \text{ km/s}$$

$$T \simeq 28 \text{ days}$$

Motion of Planets in Solar System. Kepler's Laws

Motion in a Central Field

Recall: a force $\vec{F}$ is called **central** if $\vec{F}(\vec{r}) = f(r)\vec{r}$. Hence, for a central force

$$\vec{r} \times \vec{F} = \vec{r} \times f(r)\vec{r} = 0.$$

On the other hand,

$$\vec{r} \times \vec{F} = \vec{\tau} = \frac{d\vec{L}}{dt}.$$

Eventually, for motion in the field of a central force,

$$\boxed{\frac{d\vec{L}}{dt} = 0}.$$

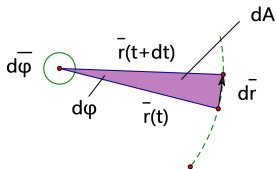
Conclusion

For motion in the field of a central force, $\vec{L} = \text{const}$ (both magnitude and direction are constant).

Note. Constant direction of $\vec{L} \implies$ motion in a plane.

Aerial Velocity

For planer motion, the **aerial velocity** may be defined



The surface area swept by $\bar{r}$ over the time dt is $dA = \left| \frac{1}{2} \bar{r} \times d\bar{r} \right|$ and the rate of change of that area

$$\frac{dA}{dt} = \frac{1}{2} \left| \bar{r} \times \frac{d\bar{r}}{dt} \right| = \frac{1}{2} |\bar{r} \times \bar{v}|.$$

Aerial velocity vector (direction — right-hand rule)

$$\boxed{\bar{\sigma} = \frac{1}{2}(\bar{r} \times \bar{v})} \quad (\text{direction same as } d\bar{\varphi})$$

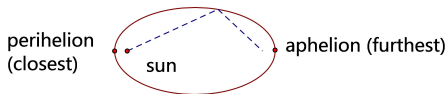
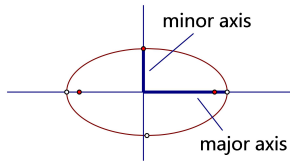
Recall: $\bar{L} = \bar{r} \times \bar{p} = \underbrace{\bar{r} \times m\bar{v}}_{=\bar{\sigma} \cdot 2m}$. Hence $\bar{L} = \text{const} \Leftrightarrow \bar{\sigma} = \text{const}$.

Consequently, for motion in a central force field $\bar{\sigma} = \text{const}$.

Kepler's Laws of Planetary Motion

- ① Each planet moves in an elliptical orbit with the Sun at one of the focal points.
- ② The line from the Sun to a given planet sweeps out equal areas in equal times.
- ③ The period of motion of a planet is proportional to the 3/2-power of the major axis length of its orbit

$$T = \frac{2\pi a^{3/2}}{\sqrt{GM_{\text{Sun}}}}.$$



Note. Precisely, motion of both the Sun and the planet is about their center of mass (in practice, the center of mass is very close to the center of the Sun).